



Identifying Causal Relationships and Response Strategies for Stakeholder Risks in Construction Projects

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Abstract: Owners, contractors, and consultants are among the stakeholders most vulnerable to high risk due to their intricate interactions over project phases. This study aims to identify and categorize the primary and secondary risk factors associated with these three parties and to examine the causal relationships among them to inform risk mitigation. A hybrid approach combining fuzzy theory and the Decision-Making Trial and Evaluation Laboratory (DEMATEL) method was adopted to analyze 32 secondary risk factors grouped under 3 primary risk factors. Nine experts, each with at least fifteen-year experience in risk management in engineering organizations, were consulted to identify and examine the risk factors. Based on the fuzzy DEMATEL (FD) model, risks were analyzed through a weighted impact matrix, and all risks were prioritized by identifying the causes leading to their effects. Owner risk (Degree of Dispatching (D) – Degree of Receiving (R) = +0.739) and contractor risk ($D - R = +0.108$) were classified as causes, whereas consultant risk ($D - R = -0.84$) was classified as an effect. The risk factors were coded RC1–RC32. From the consultant side, the second most prominent sub-factor ($D + R = 2.635$) was regulatory non-compliance (RC15), which affected approval timelines and coordination with the consultant. As regards the owner, poor communication (RC5, $D + R = 5.756$) was a causal sub-factor. For the contractor, regulatory non-compliance (RC26, $D + R = 2.690$) was the most prominent causal sub-factor. The analysis revealed divergent risk perceptions among stakeholders, thus highlighting the importance of a collaborative risk management framework.

Keywords: Construction project risk; Stakeholder risk; Fuzzy DEMATEL; Causal analysis; Risk mitigation

1 Introduction

Stakeholders' interrelationship in construction projects is a complex relationship, and traditional project management practices have created numerous risks throughout the project lifecycle. Consequently, the integration of modern techniques into risk analysis and assessment has become crucial. These types of projects are surrounded by significant uncertainty, technical, organizational, financial, and environmental, if left unmanaged, leads to delays, increased costs, and declining quality [1, 2]. The construction sector is a major driver of global economic development, despite the inherent risks of its projects. According to the Project Management Institute (PMI), traditional risk management practices, the primary reason for the failure of 14% of construction projects, are shown to be ineffective [3, 4]. While modern risk management is essential to mitigate these risks, traditional risk assessment tools often overlook the complex networks and inherent ambiguities of construction risks [5]. Project stakeholders typically face a variety of risks, such as disagreements between consultants, design constraints, and revision requests. Owners sometimes face challenges related to regulatory compliance and the project's financial viability. Construction companies also suffer from shortages of skilled labor and logistical disruptions. [6]. For example, the owner may incur financial losses, and the consultant's reputation may be damaged if the contractor is late in completing the project (e.g., due to a shortage of materials). Traditional risk matrices or probabilistic models often oversimplify these relationships, leading to suboptimal mitigation strategies [7]. Recent developments in Multi-Criteria Decision-Making (MCDM) methods promise to address these problems. Decision-Making Trial and Evaluation Laboratory (DEMATEL) is a structural modeling approach for studying cause-and-effect relationships between interacting systems [8]. Combined with fuzzy

set theory to quantify linguistic uncertainty, the fuzzy DEMATEL (FD) model is very well suited for construction risk analysis because expert judgments can often be wrong [9, 10]. Although DEMATEL methodology has been applied in both supply chain risk management and occupational safety on construction sites risk factors, its application to assess multi-stakeholder risk interactions in the construction sector [11, 12]. The extensive range of potential risk factors related to stakeholder risk interaction should be examined comprehensively and systematically [13]. Published studies have demonstrated the feasibility of DEMATEL for project risk analysis [14] and the value of fuzzy logic for processing expert knowledge [15]. These findings provided a basis for the proposed model. This research focused on three main stakeholders in project execution (owners, consultants, and contractors) and their associated sub-risks. An approach integrating fuzzy logic and DEMATEL was applied to provide an analytical framework that objectively and effectively addressed the ambiguity in the causal relationships among the three stakeholders, thereby supporting risk reduction and mitigation. Recent work has demonstrated the value of fuzzy MCDM in construction [16, 17]. However, the published studies did not deal with risk interactions among project owners, consultants, and contractors, having taken into consideration inter-relationship. This research bridged that gap by proposing a fuzzy causal analysis model for the three principal stakeholders in construction projects.

Despite adopting the same basic FD approach, the current work differed from previously published papers in three main issues that alter the subject of analysis and its implications. Initially, the three project stakeholders, i.e., owner, consultant, and contractor, were treated as a single 3×3 causal matrix node, replacing the risk type categories as the unit of analysis. The model did not place causes under stakeholders in a hierarchical sequence; instead, it illustrated risk transmission across stakeholders. Second, the expert team was purposefully chosen to be balanced, representing all three stakeholders rather than just the owner's side, and the 32 sub-factors were classified by the responsible stakeholders rather than by risk type. Third, the analysis demonstrated a stakeholder-specific response strategy framework that connected each causal factor to the responsible party and to tangible governance/digital tools, as well as a directional risk transmission chain (owner $\rightarrow$ contractor $\rightarrow$ consultant) that was not discussed in the previous studies. These variations are summed up in Table 1. The present study was also based on an entirely new expert panel and a novel dataset collected specifically for this work: none of the 9 admissible questionnaires analyzed here were used in the earlier study, and the panel was balanced across the three stakeholder groups instead of being drawn from the owner's side alone. Beyond the change of risk categories, the substantive difference was therefore the unit of analysis itself: the previous study ranked risk categories inside a single party, whereas the present study treated the three parties as interacting nodes and quantified the direction and strength of risk transmission between them.

Table 1. Comparison between the authors' previous and current studies

Main Differences	Previous Studies	Current Study
Research aim/objectives	Ranking of owner-side risk categories based on fuzzy Decision-Making Trial and Evaluation Laboratory (DEMATEL) under a single-stakeholder	Model owner-consultant-contractor as one interacting system to keep track of how risk is transmitted across parties
Unit of analysis/risk items	8 major risk types; 46 sub-factors grouped by risk type	3 stakeholders as matrix nodes; 32 sub-factors by the responsible party
Expert sample and panel composition	9 specialists from owner's side only	9 specialists (owner, consultant, and contractor purposively balanced 12 approached and 3 excluded)
Matrix and threshold values	8×8 matrix; threshold = 0.195	3×3 matrix of stakeholders; threshold = 1.173
Main output	Top causative category was Force Majeure/External Risks	Causal factors: Owner ($D - R = +0.73$) and Contractor ($D - R = +0.108$); affected factor: Consultant ($D - R = -0.84$)
Conclusions and recommendations	Causality analysis in a single party's risk set No cross-party transmission mechanism	Identifies an Owner $\rightarrow$ Contractor $\rightarrow$ Consultant risk-transmission pathway and adds a stakeholder-specific response-strategy table that is absent from the previous study

By doing so, this research contributes to both academic and practical understanding. Academically, the research encouraged the use of a fuzzy hybrid methodology for MCDM in construction management, providing a general platform applicable to the analysis of multi-stakeholder systems. Practically, the research supported project managers by offering a structured methodology for visualizing risk networks, allocating resources efficiently, and improving stakeholders' collaboration. The objectives of this research are as follows: (1) identify and classify the primary risk

factors associated with owners, consultants, and contractors that should be considered through a literature review and expert opinions to validate the findings; (2) apply fuzzy relationship analysis to calculate the causal relationships between these risk factors, including determining their significance; and (3) calculate the weights of primary criteria based on their relative significance compared to secondary criteria. Ultimately, the research aims to bridge a gap by improving targeted decisions through the development of a risk analysis theory related to construction projects and proposing an integrated quantitative approach that is closer to reality in the construction process, taking into account the dynamic interactive relationship between the main parties involved in this process: owner, consultant, and contractor.

2 Methodology

This section presents the research questions (RQs), the procedure for identifying owner, consultant, and contractor, risk factors from the literature and screening them with an expert panel, as well as the analytical method. It then describes the FD procedure used to determine the causal relationships among these factors. The technique of FD is prominent and the standard steps are presented with their references. The complete set of intermediate matrices is narrated in Section 3 so that the analysis could be reproduced from this paper. The application-specific details are also described to enhance replicability.

2.1 Research Questions

To address the research gaps, three RQs were formulated:

RQ1: What risk variables from the owner, consultant, and contractor operate as upstream generators of risk for the other parties and which variables act largely as downstream receivers of risk communicated from them?

RQ2: Which risk factors carry the greatest combined significance and impact on overall project performance, and how should this ranking guide the allocation of mitigation resources across stakeholders?

RQ3: How are the identification and efficacy of suggested response methods impacted by the interconnections and causal links between various stakeholder risks?

2.2 Research Methodology

The population of the study included twelve experts who were working on construction projects. According to measures of risk factors, the classical procedure and FD approach were employed to obtain an overall view by means of which these study points were established to categorize risk factors as main or subsidiary factors for analyzing internal causes among main and other specific criteria (weight valuation). The suggested procedure for analyzing and evaluating the primary sub-factors in the model of evaluating risk factors using the FD analysis method is depicted in Figure 1.

Twelve experts with more than fifteen-year of professional experience in risk management or construction project management were recruited through purposive sampling. The experts had worked in the roles of owner, consultant, or contractor in the private or public sector with high educational degree. They completed a pairwise-comparison questionnaire with full admissibility checking. The returned questionnaires were checked for eligibility, completeness, and internal consistency. Entries validity means a clear and single term from five-point scale, while the completeness purpose targeted one linguistic term from direct-relation matrix in each pairwise cell. Internal consistency required the choice of “No influence” to be placed in the diagonal cells for the non reversed row and column roles matrix in order to confirm parties not influencing their choice in the matrix.

Three returned questionnaires were excluded because they did not meet the eligibility criteria; the analysis therefore used only nine eligible responses.

2.3 Literature Review

Reducing delays, cost overruns, and subpar quality in construction projects require effective risk management. Conventional approaches to risk assessment frequently depend on qualitative studies, which are imprecise in addressing unexplicit or interrelated risk variables. Fuzzy logic and recent developments in MCDM methodologies, such as DEMATEL method, have enhanced the measurement of intricate risk interconnections. With an emphasis on the functions of owners, consultants, and contractors as well as the use of FD, this review of the literature compiled recent research on risk factor assessment in building projects. The fragmented nature of stakeholder risk identification has been highlighted by prior research. The contractors are more susceptible to supply chain and operational disruptions, whereas owners are primarily exposed to contractual and financial risks [6]. In their capacity as intermediaries, consultants usually have to cope with faulty designs and inadequate communication [18]. However, these studies usually ignore linkages in favor of focusing on risks separately. DEMATEL is a helpful tool for mapping the causal relationships between components.

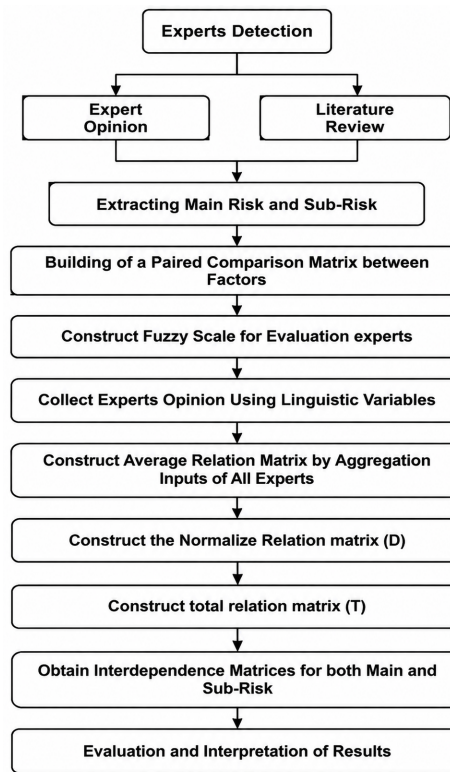


Figure 1. Methodology of risk factor assessment and examination

For instance, DEMATEL was preferred to evaluate risks in infrastructural projects though it was acknowledged that the linguistic challenges associated with expert judgment was unable to be completely addressed [19]. Fuzzy logic integration tackles these constraints by transforming subjective expert judgments into quantifiable truths [20]. Fuzzy sets enlarge the reliability of DEMATEL in scenarios with inadequate knowledge, especially in large-scale projects [12]. Recent study has demonstrated the value of FD in sustainability risk assessment and validated its ability to categorize components based on their impact and dependence [21]. Nevertheless, few studies have completely addressed the triple-stakeholder perspective (owner, consultant, and contractor) using this hybrid technique. The FD model outperformed traditional risk assessment methods in identifying the fundamental causes of construction industry delays [11]. The adaptability of this method was clarified by representing the dynamic interdependence of hazards in megaprojects using Bayesian networks in combination with the FD model [22, 23]. Despite recent improvements, there are still gaps in the systematic assessment of how risks propagate across stakeholder borders throughout project implementation. Table 2 represents aspects of FD methods adopted in previous studies.

Consultants are exposed to technical and contractual risks, including inefficient design, ambiguous definition of scope, and documentation errors, leading to rework and disputes [1, 27]. Contractors continuously face operational risks, such as shortage of labor, equipment malfunctions, and safety hazards, which impair productivity and increase accident-related liabilities [28, 29]. These risks collectively contribute to delays, budget overruns, and decreased quality, undermining stakeholders' confidence and project viability. Robust contracts, proactive screening, and collaborative planning are essential for effective risk mitigation that ensure stakeholder accountability and objective alignment [30]. To do so, it is necessary to specify which criteria in the relevant sections of the contract to be applied to the contractor and to recognize that these criteria do not necessarily lead to choices that are rational for both parties, since what is most rational from the contractor's viewpoint might be irrational from the owner's viewpoint [6].

Secondary risk factors were identified by the researcher and past experiences in the field of engineering project management and implementation through brainstorming (BS). They were identified from various previous studies, from research recently published in peer-reviewed scientific journals, and from reports of the Ministry of Planning, in addition to the main engineers syndicate. After that, expert interviews (EI) were continuously conducted with project management experts, contractors, and consultants to identify other secondary risk factors. Finally, the secondary risk factors were classified into primary factors through interviews and by taking the opinions of experts. Accordingly, the purpose of this section is to review owner decision procedures based on his criteria, list these main decision criteria with their associated sub-criteria, and establish a compilation of related risks. In Table 3, the factors are explained along with the detection methodology or reference that some secondary/sub-criteria are based on BS techniques and EI in construction project management.

Table 2. Previous studies on risk factors

Researchers	Focus Area	Major Conclusions
Seker and Zavadskas [11]	Construction sites occupational risks	Occupational hazards causal factors evaluated by FD using a cause-and-effect diagram that supports targeted safety measures; results of robustness confirmed by sensitivity analysis.
Tiwari et al. [12]	Road construction critical risk factors	Separation of critical risk factor analysis using FD from cause-and-effect for risk prioritization.
Jalhoom and Mahjoob [16]	Construction project risks	PMI risk breakdown structure with fuzzy grey DEMATEL approach adopted for project risk causes prioritization based on expert judgments.
Zhu et al. [24]	Emergency management optimization	Organizational structure and government leadership are dominant causal factors in emergency response systems.
Zhong and Zhang [23]	Prefabrication of building projects schedule risk	Hierarchical key schedule, risk factors relationships and identification using DEMATEL-ISM combined Bayesian networks.
Rostamnezhad et al. [25]	Highway projects social sustainability	Social sustainability factors: qualification and interrelationships that system dynamic modeled.
Yu and Ma [26]	Mega-infrastructure EPC projects supply chain	Supply chain risk factors causality and importance measurement using AHP-FD model.
Ahmadi et al. [22]	Process industry safety	Combines FD with Bayesian networks to prioritize organizational factors in dynamic risk updates.

Note: FD = fuzzy DEMATEL; PMI = Project Management Institute; DEMATEL = Decision-Making Trial and Evaluation Laboratory; ISM = Interpretive Structural Modeling; AHP = Analytic Hierarchy Process; EPC = Engineering, Procurement, and Construction.

Table 3. Detected risk factors

Risk Category	Risk Factor	Code	Description	Source
Owner risks	Financial instability	RC1	Insufficient budget allocation, delayed payments, or funding shortages.	[31], EI, BS
	Scope changes	RC2	Frequent or poorly defined alterations to project scope post-approval.	[32], EI, BS
	Inadequate planning	RC3	Lack of feasibility studies, unclear objectives, or incomplete design briefs. Unrealistic schedule “overly aggressive timelines that ignore construction realities or risks”.	[33], EI, BS
	Contractual disputes	RC4	Ambiguous contract terms, unrealistic penalties, or adversarial relationships.	[34]
	Poor communication	RC5	Ineffective stakeholders’ engagement or fragmented decision-making processes.	[30]
	Owner’s lack of experience	RC6	Limited technical expertise in construction management or procurement.	EI, BS
	Site readiness issues	RC7	Inadequate site preparation (e.g., land acquisition delays, geotechnical risks).	[35]
	Stakeholders’ conflicts	RC8	Delays due to objections from community groups, investors, or government bodies. Poor stakeholders’ engagement escalates this risk.	EI, BS
	Site condition uncertainties	RC9	Unforeseen ground conditions (e.g., soil instability, buried utilities) that increase costs or require design modifications. Owners often assume this risk unless transferred via contracts.	EI, BS
	External risks	RC10	Macro risks like economic downturns, political instability, natural disasters, or supply chain disruptions (e.g., material shortages). Owners may mitigate via insurance or contingency planning.	[36, 37], BS

Consultant risks	Design inefficiencies/ flaws	RC11	Errors or omissions in design documents, leading to rework, delays, or cost overruns. Consultants may face liability for unresolved design conflicts or outdated specifications.	EI, BS
	Scope creep	RC12	Uncontrolled changes to project scope without formal approval, often due to ambiguous client requirements or poor change management. Consultants must enforce scope documentation and approval processes.	[32], EI, BS
	Contractual disputes	RC13	Ambiguities in contract terms, disagreements over deliverables, or failure to meet obligations. Consultants may face legal challenges if contracts lack clarity on roles, timelines, or payment terms.	EI, BS
	Inadequate communication	RC14	Miscommunication between consultants, contractors, and clients, resulting in errors or delays. Poorly defined communication protocols exacerbate this risk.	[30]
	Regulatory non-compliance	RC15	Failure to adhere to local building codes, environmental regulations, or safety standards. Consultants are responsible for ensuring designs can meet legal requirements.	[38]
	Technological obsolescence	RC16	Use of outdated tools or methods, leading to inefficiencies. Consultants must adopt modern tools (e.g., BIM) to remain competitive and avoid design clashes.	BS
	Delayed approvals	RC17	Delays in obtaining permits or stakeholders' approvals, are often due to incomplete documentation or bureaucratic hurdles. Consultants must streamline approval workflows.	[28], EI
	Conflict resolution failures	RC18	Inability to resolve disputes between stakeholders promptly, leading to project stagnation. Consultants should establish formal conflict resolution boards.	[39]
	Inaccurate documentation	RC19	Errors in technical drawings, specifications, or reports, causing rework or legal disputes. Consultants must implement rigorous document review processes.	[18]
	Resource allocation errors	RC20	Poor planning of human or material resources, leading to bottlenecks. Consultants must use advanced resource management tools to optimize allocations.	[1], BS
Contractor risks	Payment delays	RC21	Delays in receiving client payments, leading to cash flow disruptions.	[40, 41], EI
	Cost overruns	RC22	Exceeding budget due to inaccurate estimates or unforeseen expenses.	[42], BS
	Contractual disputes	RC23	Disagreements over contract terms, causing delays or litigation.	[39], EI
	Safety non-compliance	RC24	Failure to adhere to safety regulations, resulting in accidents or penalties.	[29]
	Material and labor shortages	RC25	The unavailability of materials and an insufficient skilled workforce are affecting timelines.	[28]
	Regulatory non-compliance	RC26	Failing to comply with legal and regulatory requirements could result in fines or stoppages.	[38]
	Technology adoption failure	RC27	Ineffective implementation of new tools or systems.	[43]
	Subcontractor default	RC28	Subcontractor failure to deliver, causing delays.	[28], BS
	Inflation/price fluctuations	RC29	Economic factors increasing material/labor costs.	[1]

Unforeseen site conditions	RC30	Unexpected ground/environmental issues requiring changes in the design.	[35], EI, BS
Equipment failure	RC31	Machinery breakdowns halting progress.	[44]
Permit delays	RC32	Administrative delays in obtaining approvals.	[45], EI

Note: BIM = Building Information Modeling; BS = brainstorming; EI = expert interviews.

2.4 Fuzzy DEMATEL Technique

Causal linkages and interdependencies among elements of complex systems could be identified using FD, a matrix-based structural modeling method [8]. FD extends DEMATEL by transforming qualitative linguistic terms using fuzzy numbers to obtain subjective expert judgment to handle linguistic impressions ambiguity [10]. Many types of risk like delays, defects, and cost overrun risks have nonlinear intercorrelation and depend on dynamic variables including financial limitations, interruptions of supply chain, and design problems that make FD capability valuable in dealing with these issues [5]. DEMATEL supports important risk prioritization (e.g., contractual uncertainty, staff shortage) and enhancement of focused mitigation measures under a condition of separating driving factors from dependent ones [46]. The soundness of judgment under application uncertainty like sustainable project success factors assessment could be integrated by FD model [21]. FD integration with complementary modelling techniques, like system dynamics, could be supporting resource allocation and harmonizing strategic goals with project management decisions [25]. It is compatible to handle the complexity of contemporary construction projects with both flexible, complementary qualitative methods and rigorous quantitative methods [47].

3 Statistical Analysis and Interpretation of Fuzzy DEMATEL Results

There were two primary parts in the questionnaire. As described below, the goal of this data analysis phase facilitates easy understanding of the questionnaire results.

3.1 Part One: Personal Information

Before the implementation of this method, building project risk criteria and sub-criteria were extracted from the viewpoint of nine experts in the employer, public, and private sectors. Twelve professionals took part in the survey, but the survey forms of three experts were excluded or cancelled due to incomplete responses, inaccuracies, or lack of clarity in their answers as described in Section 2.2. Therefore, nine experts were selected for the survey, and six were men while three were women. Table 4 presents the general information of the expert sample that gives a considerable description of it. The subjects were recruited by purposive sampling. The experts included in this study had different education levels, more than 15 years of professional experience, and similar professional responsibilities. The aggregated direct-relation matrix generated based on expert judgments is shown in Table 5.

An aggregated individual assessment process was performed by calculating each expert fuzzy linguistic arithmetic mean of each pair of criteria (as shown in step 1 below). All nine experts have an equal weight with a balanced combination of three members for each role (owner, consultant, and contractor) as shown in Table 4.

Table 4. General information for specialists (experts)

Features	Type	Percentage
Role in the project	Owners	33%
	Consultants	33%
	Contractors	33%
Sex	Man	66%
	Woman	33%
Years of experience	Above 30	33%
	22–30	33%
	17–21	33%

Note: $n = 9$ experts. Percentages were rounded to represent the proportion of the nine admissible responses.

Table 5. Direct-relation matrix

Risk Owner	Owner	Consultant	Contractor
Owner	0.000, 0.000, 0.000	0.750, 1.000, 1.000	0.750, 1.000, 1.000
Consultant	0.250, 0.500, 0.750	0.000, 0.000, 0.000	0.250, 0.500, 0.750
Contractor	0.583, 0.833, 1.000	0.417, 0.667, 0.830	0.000, 0.000, 0.000

Note: Each entry lists the low (L), middle (M), and upper (U) values of a triangular fuzzy number (TFN) in that order.

3.2 Part Two: Analysis and Assessment of Principal and Sub-Risk Factor

3.2.1 Risk matrix processing

To achieve the failure diagnosis model, the researcher compared several major criteria with a series of key steps as illustrated below.

Step 1. Generate fuzzy direct-relation matrix: based on the n -criterion relationship model, the first step is to obtain an $n \times n$ matrix. The scalar product between row and column in this matrix could be formulated as a fuzzy number. If the judgments of multiple experts are employed, everybody has to fill out the matrix. The direct-relation matrix z is generated by taking the arithmetic average of all experts' opinions [10]. The language phrases accessible to the experts and the triangular fuzzy number (TFN) allocated to each of them are shown in Table 6.

$$z = \begin{bmatrix} 0 & \cdots & \tilde{z}_{n1} \\ \vdots & \ddots & \vdots \\ \tilde{z}_{1n} & \cdots & 0 \end{bmatrix} \quad (1)$$

where, z represents the initial fuzzy direct-relation matrix of size $n \times n$.

Table 6. Comparing evaluation criteria pairwise with a fuzzy scale

Code	Linguistic Terms	Symbols of Linguistic Terms	Fuzzy Numbers		
			L	M	U
1	No influence	No	0		0.25
2	Very low influence	VL	0	0.25	0.5
3	Low influence	L	0.25	0.5	0.75
4	High influence	H	0.5	0.75	1
5	Very high influence	VH	0.75	1	1

Note: L, M, and U denote the respective lower, middle, and upper values of a triangular fuzzy number (TFN), as defined in Eq. (1).

Step 2. Normalize the fuzzy direct-relation matrix: the normalized fuzzy direct-relation matrix $\tilde{X} = [\tilde{x}_{ij}]$ was obtained using Eqs. (2)–(3), and the results are shown in Table 7.

$$\tilde{x}_{ij} = \frac{\tilde{z}_{ij}}{r} = \left(\frac{l_{ij}}{r}, \frac{m_{ij}}{r}, \frac{u_{ij}}{r} \right) \quad (2)$$

where,

$$r = \max_{i,j} \left\{ \max_i \sum_{j=1}^n u_{ij}, \max_j \sum_{i=1}^n u_{ij} \right\}, \quad i, j \in \{1, 2, 3, \dots, n\} \quad (3)$$

where:

$\tilde{z}_{ij}$ denotes the fuzzy influence of factor i on factor j , expressed as a TFN (l_{ij}, m_{ij}, u_{ij}) ;

n is the total number of risk factors ($i, j \in \{1, 2, \dots, n\}$);

x_{ij} is the normalized fuzzy direct relation between factor i and factor j ;

r serves as the normalization constant, calculated as the maximum sums of the upper bounds;

u_{ij} is to ensure all fuzzy values remain within the $[0, 1]$ interval; and

l_{ij} , m_{ij} , and u_{ij} represent the respective lower, middle, and upper bounds of the TFNs in the direct-relation matrix.

Table 7. Normalized fuzzy direct-relation matrix

Risk Owner	Owner	Consultant	Contractor
Owner	0.000,0.000,0.000	0.375,0.500,0.500	0.375,0.500,0.500
Consultant	0.125,0.250,0.370	0.000,0.000,0.000	0.125,0.250,0.370
Contractor	0.292,0.417,0.500	0.209,0.330,0.417	0.000,0.000,0.000

Step 3. Calculate the fuzzy total-relation matrix: the fuzzy total-relation matrix could be calculated by the following formula:

$$\tilde{T} = \lim_{k \rightarrow +\infty} (\tilde{x}^1 \oplus \tilde{x}^2 \oplus \dots \oplus \tilde{x}^k) \quad (4)$$

where, $\tilde{T}$ signifies the fuzzy total-relation matrix; I is the identity matrix of dimension $n \times n$; and k represents the power index denoting the sequence of direct and indirect influences ($k \rightarrow \infty$).

$\tilde{x}^1, \tilde{x}^2, \dots, \tilde{x}^k$ are the consecutive powers of the normalized fuzzy direct-relation matrix $\tilde{x}$ (see Table 7), which reflects the indirect influence channels of length 1, 2, $\dots$, k accordingly. Their fuzzy sum ($\oplus$) converges to the fuzzy total-relation matrix $\tilde{T}$ as $k \rightarrow \infty$. The identity matrix mentioned above was utilized in Eqs. (5)–(7) below, not in Eq. (4) itself.

The operator $\oplus$ denotes fuzzy addition for TFNs. If each element of the fuzzy total-relation matrix is expressed as , it can be calculated as follows:

$$[l''_{ij}] = x_l \times (I - x_l)^{-1} \quad (5)$$

$$[m''_{ij}] = x_m \times (I - x_m)^{-1} \quad (6)$$

$$[u''_{ij}] = x_u \times (I - x_u)^{-1} \quad (7)$$

The double-prime notation used in l''_{ij} , m''_{ij} , and u''_{ij} from Eqs. (5)–(7) distinguishes the boundary values of the total-relation matrix $\tilde{T}$ from those of the initial or normalized matrices. The crisp $n \times n$ matrices x_l , x_m , and x_u in Eqs. (5)–(7) were obtained by aggregating the lower (l), middle (m), and upper (u) bounds of each element of the normalized fuzzy direct-relation matrix $\tilde{X}$ (Table 7), respectively. Each value was then applied to the matrix inverse $(I - x)^{-1}$. The elements of the fuzzy total-relation matrix were denoted in three ways throughout this derivation: $\tilde{T}$ as the matrix name in Eq. (4), $l''/m''/u''$ with a double prime here are from Eqs. (5)–(7), and $l^t/m^t/u^t$ with a superscript t are from Eqs. (8)–(11) below. All three were related to the same fuzzy total-relation matrix $\tilde{T}$ (Table 8).

The normalized matrix was multiplied by the resultant matrix after the inverse was first computed and subtracted from matrix I , which yielded the fuzzy total-relation matrix, as shown in Table 8.

Table 8. Fuzzy total-relation matrix

Risk Owner	Owner	Consultant	Contractor
Owner	0.226,0.872,2.599	0.571,1.362,3.021	0.531,1.276,2.932
Consultant	0.203,0.723,2.399	0.121,0.617,2.199	0.216,0.766,2.399
Contractor	0.400,1.021,2.799	0.400,1.106,2.843	0.200,0.787,2.465

Note: The entries of this table were reported to three decimal places, exactly as returned by the computations of Eqs. (5)–(7). It is the input to converting fuzzy data into crisp scores (CFCS) defuzzification in Step 4.

Step 4. Defuzzify into crisp values: The total-relation matrix's crisp value has been obtained using the converting fuzzy data into crisp scores (CFCS) approach as proposed by Opricovic and Tzeng [48]. The steps of CFCS method are as follows:

$$l^n_{ij} = \frac{l^t_{ij} - \min l^t_{ij}}{\Delta^{\max}_{\min}} \quad (8)$$

$$m^n_{ij} = \frac{m^t_{ij} - \min l^t_{ij}}{\Delta^{\max}_{\min}} \quad (9)$$

$$u^n_{ij} = \frac{u^t_{ij} - \min l^t_{ij}}{\Delta^{\max}_{\min}} \quad (10)$$

$$\Delta^{\max}_{\min} = \max u^t_{ij} - \min l^t_{ij} \quad (11)$$

In Eqs. (8)–(11), l^n_{ij} , m^n_{ij} , and u^n_{ij} are the normalized lower, middle, and upper bounds of the total-relation element (i, j); the superscript n indicates normalization relative to the range $\Delta^{\max}_{\min}$ as defined in Eq. (11), which is the spread between the highest upper bound and the lowest lower bound taken across all elements of the fuzzy total-relation matrix $\tilde{T}$ (Table 8). l^t_{ij} and u^t_{ij} (superscript 't') are the equivalent unnormalized total-relation limits, which are the same values as l''/u'' from Eqs. (5)–(7).

Calculating the upper and lower bounds of normalized values:

$$l_{ij}^s = \frac{m_{ij}^n}{1 + m_{ij}^n - l_{ij}^n} \quad (12)$$

$$u_{ij}^s = \frac{u_{ij}^n}{1 + u_{ij}^n - m_{ij}^n} \quad (13)$$

In Eqs. (12)–(13), l_{ij}^s and u_{ij}^s are the standardized (scaled) lower and upper limits of element (i, j) , calculated from the normalized values l_{ij}^n , m_{ij}^n , and u_{ij}^n specified above. The superscript s represents this standardization step, which is done before final defuzzification in Eq. (14).

The output of the CFCS algorithm is crisp values. After calculating total normalized crisp values, the figures are illustrated in Table 9:

$$c_{ij} = \frac{l_{ij}^s (1 - l_{ij}^s) + u_{ij}^s \times u_{ij}^s}{1 - l_{ij}^s + u_{ij}^s} \quad (14)$$

In Eq. (14), c_{ij} is the final defuzzified crisp value of the total-relation element (i, j) calculated from l_{ij}^s and u_{ij}^s . A separate symbol was adopted so that the crisp value was not confused with the normalized fuzzy direct-relation value $\tilde{x}_{ij}$ in Eq. (2) and Table 7. The final crisp entry of the total-relation matrix was then obtained as $T_{ij} = \min l^t + c_{ij} \cdot \Delta$ and these entries are reported in Table 9.

Table 9. Crisp total-relation matrix before thresholding

Risk Owner	Owner	Consultant	Contractor
Owner	1.11*	1.521	1.451
Consultant	0.98*	0.869*	1.014*
Contractor	1.253	1.32	1.037*

Note: * indicates an entry below the threshold of 1.173. These entries are set to zero in the thresholded matrix shown in Table 11 but are retained when calculating D and R from Table 9.

In Eqs. (11)–(14), three aspects of CFCS were used for the generation of numerical results in Table 9. First, the range Δ of Eq. (11) was computed for each column: for each column j of the fuzzy total-relation matrix, $\min l^t$ and $\max u^t$ were calculated over the entries of that column, and Eqs. (8)–(10) normalized the three bounds of every element of column j in relation to the range of that column. Second, $u^s = u^n / (1 + u^n - m^n)$ was calculated as the standardized upper limit of Eq. (13) in the original formulation of Opricovic and Tzeng [48]; $l^s = m^n / (1 + m^n - l^n)$ was the lower bound calculated using Eq. (12). Third, the value on the normalized $[0, 1]$ scale was given by Eq. (14), and the crisp item presented in Table 9 was restored by restoring the column's original scale: $x^* = \min l^t + x_{ij} \times \Delta$. These three conventions are shown stepwise for one element in the working example below Table 9.

The following practical example exhibits the way to handle the linguistic assessment from a single expert to the final and precise value, from the owner to the consultant.

Working example: following opinions of nine experts from linguistic evaluation to the final crisp value (Owner $\rightarrow$ Consultant). To make the computational chain of Steps 1–4 completely visible, the Owner $\rightarrow$ Consultant entry is tracked through each of the transformations it experiences below.

Step 1. Linguistic aggregation and evaluation

The nine experts appraised the effect the owner had on the consultant using the five-point scale as shown in Table 6. The panel reached the consensus on “very high influence (VH)”. This was mapped to the TFN $(l, m, u) = (0.750, 1.000, 1.000)$ using the scale. The aggregated judgment was the element-wise arithmetic mean of the nine ratings, $\tilde{z}$ (Owner $\rightarrow$ Consultant) $= (1/9) \sum_k (l_k, m_k, u_k) = (0.750, 1.000, 1.000)$, which is the value shown in the Owner row/Consultant column of Table 5. The three bounds were independently averaged one by one if the panel was not consensual.

Step 2. Normalization (Eqs. (2)–(3))

When the constant r is normalized, it is the upper bounds, maximum summation of column and row, as shown in Table 5. The owner provides $0.000 + 1.000 + 1.000 = 2.000$, that is all columns and rows maximum. Dividing the three boundaries by r yields $\tilde{x}$ (Owner $\rightarrow$ Consultant) $= (0.750/2.000, 1.000/2.000, 1.000/2.000) = (0.375, 0.500, 0.500)$, which is the equivalent item in Table 7.

Step 3. Fuzzy total-relation value (Eqs. (4)–(7))

In Table 7, form the three crisp matrices x_l , x_m , and x_u , each of which is adjusted by $x(I - x)^{-1}$. For the Owner $\rightarrow$ Consultant position, this gives $\tilde{t}$ (Owner $\rightarrow$ Consultant) $= (0.571, 1.362, 3.021)$ as in the entry of Table 8. The

middle limit has increased from 0.500 to 1.362 since the total-relation matrix also contained the indirect impact of the owner on the consultant, which is conveyed via the contractor.

Step 4. Defuzzification (Eqs. (8)–(14))

In the Consultant column of Table 8, the lowest lower limit is $\min l^t = 0.121$ (Consultant $\rightarrow$ Consultant) and the biggest upper bound is $\max u^t = 3.021$ (Owner $\rightarrow$ Consultant). Therefore, using Eq. (11) $\Delta = 3.021 - 0.121 = 2.900$, the normalized limits from Eqs. (8)–(10) are $l^n = (0.571 - 0.121)/2.900 = 0.155$, $m^n = (1.362 - 0.121)/2.900 = 0.428$ and $u^n = (3.021 - 0.121)/2.900 = 1.000$. The standardized boundaries are given by Eqs. (12)–(13) as $l^s = 0.428/(1 + 0.428 - 0.155) = 0.336$ and $u^s = 1.000/(1 + 1.000 - 0.428) = 0.636$. Eq. (14) then gives $c_{ij} = [0.336 \times (1 - 0.336) + 0.636 \times 0.636]/(1 - 0.336 + 0.636) = 0.628/1.300 = 0.48$. Restoring the column's original scale gives $T_{ij} = \min l^t + c_{ij} \Delta = 0.121 + 0.483 \times 2.900 = 1.521$, which is precisely the crisp value for Owner $\rightarrow$ Consultant reported in Table 9.

Use of the value in the rest of the analysis. The value 1.521 is above the threshold of 1.173 as defined in Step 5, so the Owner $\rightarrow$ Consultant link is in the network relationship map. The same value is in the Owner row sum $D = 1.110 + 1.521 + 1.451 = 4.083$ and the Consultant column sum $R = 1.521 + 0.869 + 1.320 = 3.710$. This gives the Cause/Effect classification. The remaining cells of the matrix were each subjected to precisely the same five steps.

Step 5. Set threshold value When creating the internal relations matrix, a threshold parameter must be defined to discriminate between different relationships. As a result, a network relationship map was made, and incomplete relations were disregarded. Network relationship map included relations, whose matrix T -values were greater than a particular threshold. The average value of the T matrix could be adopted to determine the threshold for relations. Given a threshold intensity, without considering the previously described causal relation, the elements in the T matrix whose values were less than the threshold intensity were set to zero. The study's cutoff value was 1.173. The aforementioned causal relationship was not taken into consideration since all items in matrix T below 1.173 were set to zero.

Threshold sensitivity: Since D and R are the row/column sums of the complete and un-thresholded total-relation matrix (Table 9), the Cause/Effect classification of each stakeholder is mathematically independent of the threshold value: Owner ($D - R = +0.739$), Contractor ($D - R = +0.108$), and Consultant ($D - R = -0.84$) all have the same sign and therefore the same identification at every threshold. What the threshold does decide is which of the six cross-party linkages are shown in the network relationship map. Table 10 sweeps this criterion across a range that includes the mean-based cutoff employed in our investigation (1.173).

The sweep validates the robustness of the presented structure. The two owner-driven links survive every cutoff from 0.90 to 1.45. The consultant has no outbound linkages at any threshold above 1.014; therefore, its position as a pure receiver of influence is maintained throughout the entire feasible range. The map is stable and interpretable in the range approximately 0.90–1.45: below 0.90, all pairs are connected, and the diagram loses its discriminating power, while at 1.55, the cutoff exceeds the largest entry of the matrix (1.521), and no linkage survives at all, fixing the practical upper limit of the criterion. Thus, the mean-based limit of 1.173 employed in this study is well within this steady range. It is vital to specify exactly what does and does not change over the sweep. The number of cross-party links kept in the network relationship map does fluctuate with the threshold, from six at $\theta = 0.90$ to five at $\theta = 1.00$, four at the study's cutoff of $\theta = 1.173$, three at $\theta = 1.30$, two at $\theta = 1.45$, and none at $\theta = 1.55$. Conversely, the cause/effect classification for the three stakeholders is the same for all thresholds tested, with Owner and Contractor as causes and Consultant as an effect. This is because D and R are derived from the full and un-thresholded crisp matrix in Table 9 and therefore are not affected by the cutoff. The interpretations given in the next sections are based on that categorization and on the four links maintained at $\theta = 1.173$, and none of them is changed elsewhere throughout the stable range investigated.

Across this spectrum, there are two solid results. First, the Owner $\rightarrow$ Consultant and Owner $\rightarrow$ Contractor linkages persist at all thresholds up to 1.45, representing the owner's position as the major upstream driver, yet not an artifact of the particular cutoff selected. Second, the Consultant never has a surviving outgoing link once the threshold exceeds 1.014; its two outgoing values (Consultant $\rightarrow$ Owner = 0.98, Consultant $\rightarrow$ Contractor = 1.014) are the two smallest of the six off-diagonal entries so the consultant's status as a pure recipient is stable and well beyond the study's chosen cutoff. It only breaks down at implausibly permissive thresholds below 1.0. At the main-category level, there are no factors that the transition between cause and effect at any threshold evaluated. The crisp total-relation matrix after applying the threshold $\theta = 1.173$ is presented in Table 11.

Step 6. Final output: Calculate the row sums (D) and column sums (R) of the crisp total-relation matrix T using the following formulas.

$$D = \sum_{j=1}^n T_{ij} \quad (15)$$

$$R = \sum_{i=1}^n T_{ij} \quad (16)$$

In Eq. (15)–(16), the sums of row and column influences are defined:

D (Degree of Dispatching) represents the sum of the i th row of the crisp total-relation matrix T , indicating the total influence factor i exerts on other factors.

R (Degree of Receiving) represents the sum of the j th column of the crisp total-relation matrix T , indicating the total influence factor i receives from other factors.

T_{ij} refers to the crisp value of the total relation between factor i and factor j from the matrix T obtained in Step 4.

The overall importance of factor i in the system is $D + R$ and net costs associated with factor i 's presence in the system are given by $D - R$. These values can then be computed based on D and R . The final result is shown in Table 12 below.

Table 10. Sensitivity of the network relationship map to the threshold value

Threshold (θ)	Retained Linkages	Cross-Party Linkage	Newly Excluded Linkages	Cause/Effect ($D - R$)
0.90	6	Owner $\rightarrow$ Consultant = 1.521, Owner $\rightarrow$ Contractor = 1.451, Contractor $\rightarrow$ Consultant = 1.320, Contractor $\rightarrow$ Owner = 1.253, Consultant $\rightarrow$ Contractor = 1.014, Consultant $\rightarrow$ Owner = 0.980	(6 connections found)	Owner: Cause = +0.739, Contractor: Cause = +0.108, Consultant: Effect = -0.840
1.00	5	Owner $\rightarrow$ Consultant = 1.521, Owner $\rightarrow$ Contractor = 1.451, Contractor $\rightarrow$ Consultant = 1.320, Contractor $\rightarrow$ Owner = 1.253, Consultant $\rightarrow$ Contractor = 1.014	Consultant $\rightarrow$ Owner = 0.980	Owner: Cause = +0.739, Contractor: Cause = +0.108, Consultant: Effect = -0.840.
1.173 (study cutoff, T mean)	4	Owner $\rightarrow$ Consultant = 1.521, Owner $\rightarrow$ Contractor = 1.451, Contractor $\rightarrow$ Consultant = 1.320, Contractor $\rightarrow$ Owner = 1.253	Consultant $\rightarrow$ Contractor = 1.014	Owner: Cause = +0.739, Contractor: Cause = +0.108, Consultant: Effect = -0.840
1.30	3	Owner $\rightarrow$ Consultant = 1.521, Owner $\rightarrow$ Contractor = 1.451, Contractor $\rightarrow$ Consultant = 1.320	Contractor $\rightarrow$ Owner = 1.253	Owner: Cause = +0.739, Contractor: Cause = +0.108, Consultant: Effect = -0.840
1.45	2	Owner $\rightarrow$ Consultant = 1.521, Owner $\rightarrow$ Contractor = 1.451	Contractor $\rightarrow$ Consultant = 1.320	Owner: Cause = +0.739, Contractor: Cause = +0.108, Consultant: Effect = -0.840
1.55	0	The network relationship map becomes empty	Owner $\rightarrow$ Consultant = 1.521; Owner $\rightarrow$ Contractor = 1.451	Owner: Cause = +0.739, Contractor: Cause = +0.108, Consultant: Effect = -0.840

Note: The thresholds were applied to the crisp total-relation matrix in Table 9. The six cross-party linkages are its off-diagonal entries; the diagonal self-influence entries (Owner $\rightarrow$ Owner = 1.110, Consultant $\rightarrow$ Consultant = 0.869, and Contractor $\rightarrow$ Contractor = 1.037) are not shown as arrows in the network relationship map. “Newly excluded” refers to linkages removed when the threshold increases from the preceding and lower value.

Table 11. Crisp total-relation matrix after applying the threshold ($\theta = 1.173$)

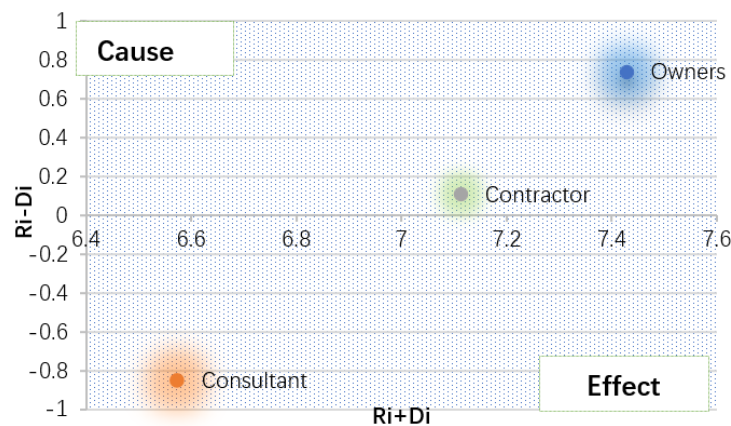
Risk Owner	Owner	Consultant	Contractor
Owner	0	1.521	1.451
Consultant	0	0	0
Contractor	1.253	1.32	0

Table 12. Final output for main risk

Main Risk	R	D	$D + R$	Rank	$D - R$	Identify
Owner	3.344	4.083	7.427	1	0.739	Cause
Consultant	3.71	2.863	6.573	3	-0.84	Effect
Contractor	3.502	3.61	7.112	2	0.108	Cause

Note: D = Degree of Dispatching; R = Degree of Receiving.

A model of the salient relationships is shown in Figure 2. This can also be represented as a diagram where the values of prominence (a factor's overall importance, $D + R$) are laid off as abscissa and those of relation (net cause/effect value of a factor, $D - R$) as the ordinate. Each factor's position and relative interaction with a coordinate point ($D + R$, $D - R$) in the Minkowski coordinates are determined by positioning the factors in those coordinates.

**Figure 2.** Final relationship among the main risks

Step 7. Interpret the results: As shown in Table 12 and Figure 2, the horizontal coordinate ($D + R$) reflects the significance of each factor in the overall system and can be used to evaluate the factors. That is, ($D + R$) represents factor i 's contribution to the total system as well as the contributions of other system factors to this factor. According to priority, the owner is in first order, and the contractor & consultant are the next rank.

The vertical/linear vector ($D - R$) signifies how strong an impact a factor has on the system. Positive $D - R$ is usually called the cause in general, and the negative $D - R$ effect. For this study, both owners and contractors are seen as independent variables; Consultants are treated as dependent variables.

From an administrative perspective, the data for Table 12 and Figure 2 were interpreted as follows:

- Owner's Risk as a Main Causal Factor ($D = 4.083$, $D - R = 0.739$)

This is due to the owner's authority to finance, modify the scope of work, and grant contractual approvals, placing them at the top of the project's decision-making hierarchy. According to the study's second hypothesis, owner's risk (such as financial delays) is known as 'project-stream root', and its consequences are transmitted through the flow of information and decisions to the consultant (via modifications to design requirements) and then to the contractor (via modifications to execution). This explains why the owner's D value is higher compared to other parties.

- Consultant's Risk as a Mainly Influenced Factor ($D - R = -0.84$)

The Consultant does not have the authority to make independent decisions regarding the project scope or financing; rather, the consultant acts as an intermediary within the contractual structure, transmitting design outputs to the contractor after receiving the owner's requirements. The consultant is more influenced by the decisions of the other two parties ($R = 3.71$) than it influences them, due to its intermediary position in the flow of information.

The exact cross-party relationships are explicit in Table 9 (the crisp total-relation matrix). After thresholding, four directed linkages are Owner $\rightarrow$ Consultant (1.521), Owner $\rightarrow$ Contractor (1.451), Contractor $\rightarrow$ Consultant (1.320) and Contractor $\rightarrow$ Owner (1.253). Consultant $\rightarrow$ Owner and Consultant $\rightarrow$ Contractor turn into zero. The asymmetry is important to the research conclusion: although the consultant receives risk from the Owner and Contractor, the consultant is not solely 'an effect' in isolation, while no meaningful risk is communicated back to either stakeholder. Each stakeholder risk cannot be assessed by single stakeholder and in a three party model with independently running for each party. A risk assessment ranks cause and effects only among the sub-factors associated with each stakeholder. The model cannot generate the relation intensity an Owner $\rightarrow$ Consultant or Contractor $\rightarrow$ Consultant that cannot be defined unless the three stakeholders are placed as interacted nodes in direct-relation matrix. The structure impact straightforward practically that consultant's internal processes are symptom

treated and targeted as mitigation resources. The exposure of consultant risk is driven by upstream choices. Paper contribution is transmission of diagonal pattern without consideration to stakeholder's number. It falsifiable causal and ranked map, that justifies Owner $\rightarrow$ Contractor interface prioritization as the primary layer for system-wide risk reduction, more than consultant's own processes. It quantifies the previously widespread but unsubstantiated assumption that problems "flow down" from owner to contractor.

3.2.2 Sub-risk criteria analysis and assessment

The results of the FD model by sub-risk criteria can also be derived by repeated Step 1–Step 7 to analyze and evaluate such sub-risk criteria. The ultimate results of risk priority model in construction projects are shown in Tables 13, 14, and 15. For each sub-risk factor, the row sum D_i and the column sum R_i of the crisp total-relation matrix were calculated, and the prominence $D_i + R_i$ and the net causal impact $D_i - R_i$ were derived from them: the first describes the strength of the element's involvement in the system, the second indicates whether the factor is a cause or an effect. As shown in Figure 3, Figure 4, and Figure 5, the final results shown in the causal relationship diagrams are as follows:

A. Owners Risk Factor (Sub-Risk)

Table 13 shows the final outcome of the owner's risk factors in construction projects. It demonstrates the relative importance of the secondary factors by finding the final ranking of the factors, as well as determining the cause and effect of each factor.

Figure 3 shows the cause-and-effect diagram based on the primary criteria. All the causal properties of the primary criteria are illustrated, where the significant relationships between the secondary criteria are depicted.

Interpretation of the results for owner risk. As illustrated and tabulated above, each factor may be assessed by the owner risk factor ranks first by measurement values in Table 12 and Figure 2, with $D + R = 7.427$ in proportion to the other criteria. At the same time, the $D - R$ indicates that occupier risk is a source of risk in a construction project. For the second criterion, results were interpreted by combining Table 13 and Figure 3 as described here.

- The horizontal vector ($D + R$) shows the extent of significance of each criterion in the whole system. In other words, ($D + R$) is the effect of factor i on the whole system and that of other system factors on it. Their importance in degrees (from the most important one) is RC1 (Financial Instability), followed by RC8, RC4, RC5, RC7, RC2, RC10, RC6, RC9 and finally RC3 is the weakest factor.

- The vertical vector ($D - R$) is the influence strength of factors described in the system. In the overall sense, a $D - R$ value > 0 is indicative of a causal variable, and a $D - R$ value < 0 is an effect. In the study, RC5 (Owner's Lack), RC6 (Poor Communication of Experience), RC7 (Site Readiness Issues), and causal variables, and in model as effect are considered RC1–RC4 and RC8.

B. Consultant Risk Factor (Sub-Risk)

Table 14 represents the final result of the analysis of consultant's secondary factors during the implementation of construction projects. It also shows the relative importance of each sub-factor, in addition to determining the causal relationship between all criteria. Figure 4 shows a cause-and-effect diagram based on main criteria. It shows all the causal features of the main criterion.

Interpretation of consultant risk results. Each factor could be assessed using the following criteria, as shown in Table 14 and Figure 4: with a $D + R$ value of 6.573, which indicates the relative relevance of the criterion in relation to the other criteria, consultant risk factors came in third place, according to the results displayed in Table 12 and Figure 2. However, the $D - R$ results show that the construction project has an impact on owner risk variables. Regarding the secondary criteria, the findings were interpreted as follows in light of Table 14 and Figure 4:

- The horizontal vector ($D + R$) corresponds to the importance of each factor in the overall system. Put another way, ($D + R$) represents the effect of factor i on the whole system and all other sets of factors in it. In terms of importance value, highest to lowest, RC13 was (2), followed by RC15/RC17/RC14/RC11/RC18/RC19 (6), then in decreasing order of degrees, RC12 and RC16, and RC20.

- The vertical $D - R$ vector indicates the effect of a factor on the system. In other words, if $D - R$ is positive, then it is a causal variable, and when $D - R$ is negative, it should be called an effect. In the present study, RC11 (Design Inefficiencies/Flaws), Communication Insufficiency, RC14, RC15, and RC16 were treated as causal variables, and RC12, RC13, RC17, RC18, RC19, and RC20 were regarded as effect variables.

C. Contractor Risk Factor (Sub-risk)

Table 15 shows the final result of the analysis of the contractor secondary factors during the implementation of construction projects. It also shows the relative importance of each sub-factor, in addition to determining the causal relationship between all criteria. Figure 5 shows a cause-and-effect diagram based on the main criteria.

Interpretation of contractor risk factors. Based on the figure and table above, the contractor risk criterion ranking reveals the second-highest priority (as per Table 12 and Figure 2) based on $D + R$ value of 7.112, showing its importance relative to the other criteria. In contrast, the $D - R$ results show that contractor risk factors are direct risks to the construction project. Regarding the secondary criteria, considering the results presented in Table 15 and those indicated in Figure 5.

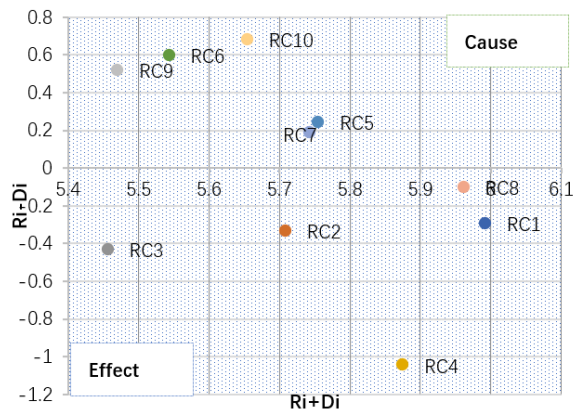


Figure 3. Final relationship among the owner's sub-risks

Note: RC1–RC10 represent the codes assigned to the secondary risk factors listed in Table 3.

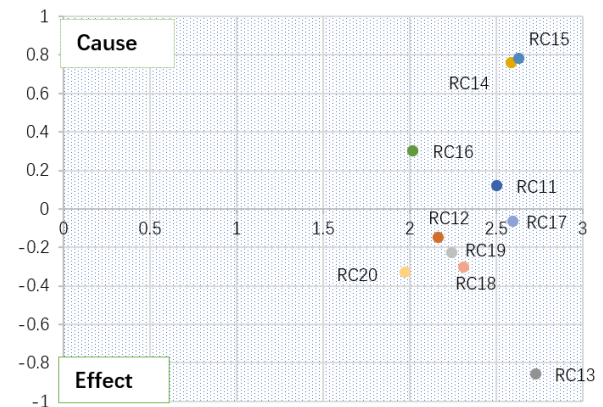


Figure 4. Final causal relationship for the sub-risk for consultant

Note: RC1–RC10 represent the codes assigned to the secondary risk factors listed in Table 3.

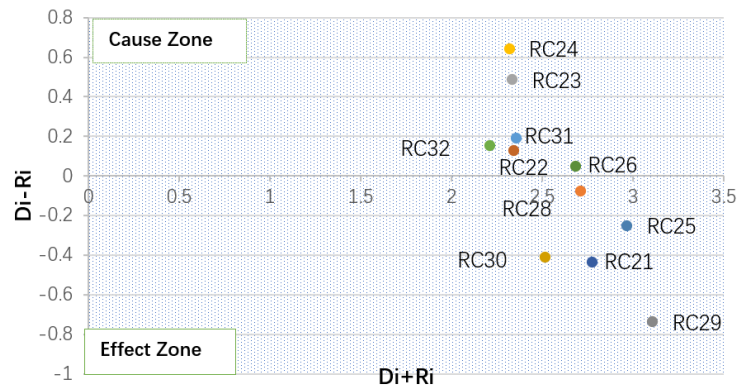


Figure 5. Final relationship for sub-risk contractor

Note: RC1–RC10 represent the codes assigned to the secondary risk factors listed in Table 3.

Table 13. Final output for owners' risk

Sub-Risk Factor	Risk Code	<i>R</i>	<i>D</i>	<i>D + R</i>	Rank	<i>D – R</i>	Identification
Financial instability	RC1	3.144	2.849	5.993	1	-0.295	Effect
Scope changes	RC2	3.021	2.687	5.709	6	-0.334	Effect
Inadequate planning	RC3	2.945	2.512	5.457	10	-0.433	Effect
Contractual disputes	RC4	3.462	2.414	5.876	3	-1.048	Effect
Poor communication	RC5	2.758	2.998	5.756	4	0.24	Cause
Owner's lack of experience	RC6	2.476	3.068	5.544	8	0.593	Cause
Site readiness issues	RC7	2.779	2.964	5.743	5	0.185	Cause
Stakeholders' conflicts	RC8	3.033	2.93	5.962	2	-0.103	Effect
Site condition Uncertainties	RC9	2.477	2.994	5.471	9	0.518	Cause
External risks	RC10	2.488	3.167	5.655	7	0.678	Cause

Note: *D* = Degree of Dispatching; *R* = Degree of Receiving.

Table 14. Final output for consultant risk

Sub-Risk Factor	Risk Code	<i>R</i>	<i>D</i>	<i>D + R</i>	Rank	<i>D - R</i>	Identification
Design inefficiencies	RC11	1.194	1.312	2.506	5	0.117	Cause
Scope creep	RC12	1.162	1.009	2.171	9	-0.154	Effect
Contractual disputes	RC13	1.796	0.935	2.731	1	-0.861	Effect
Inadequate communication	RC14	0.917	1.673	2.591	4	0.756	Cause
Regulatory non-compliance	RC15	0.928	1.707	2.635	2	0.779	Cause
Technological obsolescence	RC16	0.863	1.16	2.023	8	0.297	Cause
Delayed approvals	RC17	1.333	1.267	2.601	3	-0.066	Effect
Conflict resolution	RC18	1.312	1.006	2.317	6	-0.306	Effect
Inaccurate documentation	RC19	1.24	1.008	2.248	7	-0.232	Effect
Resource allocation errors	RC20	1.154	0.823	1.978	10	-0.331	Effect

Note: *D* = Degree of Dispatching; *R* = Degree of Receiving.

• The horizontal vector (*D + R*) shows how important each component is to the system as a whole. To put it another way, (*D + R*) shows how factor *i* affects the system as a whole, as well as how other system factors affect it. RC23 (Contractual Disputes) was ranked 1 in terms of relevance, followed by RC25, RC21, RC28, RC26, RC27, RC24, RC31, RC22, RC29, RC30, and RC32.

• The vertical vector (*D - R*) shows how much of an impact the factor has on the system. Generally speaking, an effect is represented by a negative *D - R* value and a causative variable by a positive *D - R* value. In this study, RC22, RC26, RC27, RC29, RC30, RC31 and RC32 were considered as causal variables, while RC21, RC23, RC24, RC25, and RC28 were considered as effect variables.

Table 15. Final output for contractors' risk

Sub-Risk Factor	Risk Code	<i>R</i>	<i>D</i>	<i>D + R</i>	Rank	<i>D - R</i>	Identify
Payment delays	RC21	1.611	1.17	2.781	3	-0.441	Effect
Cost overruns	RC22	1.11	1.235	2.346	8	0.125	Cause
Contractual disputes	RC23	1.927	1.186	3.113	1	-0.741	Effect
Safety non-compliance	RC24	1.467	1.053	2.52	7	-0.414	Effect
Material and labor shortages	RC25	1.613	1.358	2.972	2	-0.255	Effect
Regulatory non-compliance	RC26	1.322	1.369	2.69	5	0.047	Cause
Technology adoption failure	RC27	1.139	1.441	2.58	6	0.302	Cause
Subcontractor default	RC28	1.397	1.318	2.715	4	-0.079	Effect
Inflation/price fluctuations	RC29	0.929	1.412	2.34	10	0.483	Cause
Unforeseen site conditions	RC30	0.842	1.481	2.323	11	0.639	Cause
Equipment failure	RC31	1.088	1.274	2.362	9	0.186	Cause
Permit delays	RC32	1.034	1.182	2.216	12	0.148	Cause

Note: *D* = Degree of Dispatching; *R* = Degree of Receiving.

3.3 Interpreting the Cause-and-Effect Results in Terms of Each Party's Contractual Authority and Decision-Making Margin

Owner ($D - R = +0.739$, most important reason). Has unilateral control over funding, scope definition, and contract approvals, choices made once and are then binding on the other two parties. This is why Owner risk enters the system as the least restricted and most simply causative variable.

Contractor ($D - R = +0.108$, weak causation). Bound by the design output of the Consultant, but maintains genuine operational autonomy over means, techniques, sequencing, and subcontracting that the contract does not fully stipulate – thus the execution-side choices still create new risk rather than just propagate it.

Consultant ($D - R = -0.84$, effect). The most restrictive contractual constraint of the three: technical output is bound to the owner's brief and relevant codes/standards, leaving relatively little independent decision-making freedom—structurally a technical conduit between Owner requirements and Contractor execution, which is why it removes risk from both other parties without creating a comparable amount of its own.

3.4 Study Results From a Systems Engineering Perspective

The current study primarily presents a risk classification, and owners, consultants, and contractors are presented as interconnected subsystems within the project management system. The three points below clearly explain the risk transfer process, control procedures, and feedback mechanism.

- **Owner, Consultant, and Contractor as Interconnected Subsystems:** Instead of presenting the macro relationship matrix Table 9 and the natural resource management diagram as separate classifications for each party, the study presents them as representations of the relationships among the three parties. The difference between the outcome and the relationship ($D - R$) indicates the net influence flow between the subsystems, where D values reflect the “output” of a subsystem to the rest of the system, while R values represent its “input” from it.

- **Risk Transfer:** Table 16 (Response Strategies) provides clear examples of risk transfer between subsystems, including the transfer of financial risk to experts, the transfer of cost risk to the contractor through fixed-price contracts, and the transfer of regulatory obligations through requirements tracking systems.

- **Control Procedures and Feedback Mechanisms:** In addition, Table 16 lists the control and feedback mechanisms identified in the study findings. These mechanisms include periodic meetings and progress reports, digital information management platforms (such as BIM 360 and Aconex) for sharing real-time updates, and periodic design reviews at different stages (30%, 60%, and 90%) with independent audits. From a systems engineering perspective, these control and feedback mechanisms can help identify and correct deviations within one stakeholder subsystem, such as a consultant's design delay, before they affect another subsystem, such as the contractor.

This explicitly accounts for the risk-transfer mechanism by mapping the four surviving cross-party linkages from Table 11 as directed edges across the three subsystems, together with the control/feedback checkpoints specified in Table 16. Figure 6 illustrates the interacting subsystems of the owner, consultant, and contractor.

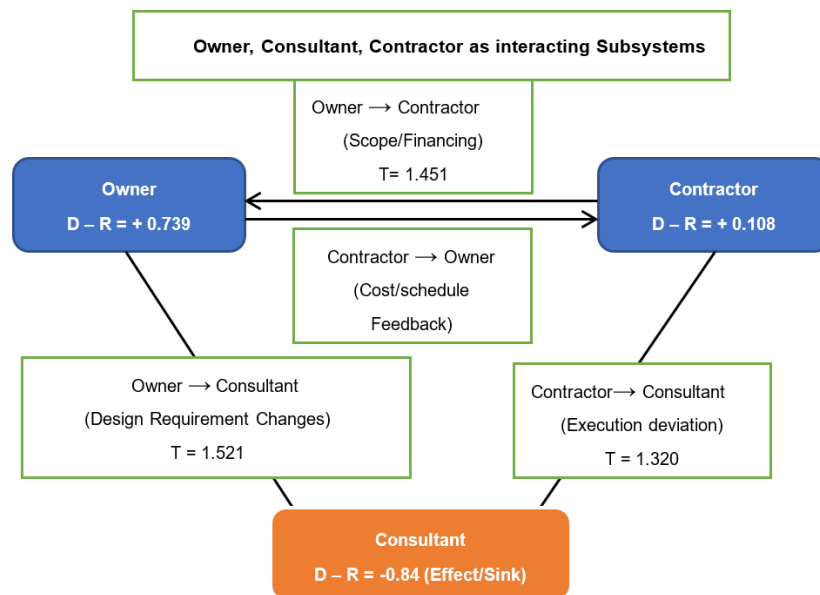


Figure 6. Risk transmission channels and control/feedback checkpoints among owner, consultant, and contractor interacting subsystems

Note: D = Degree of Dispatching; R = Degree of Receiving.

Table 16. Risk response strategies and methodologies

Risk Category	No.	Causal Factor	Key Response Strategies	Interpretation of Response Methodologies	Ref.	Expert Interviews (EI)	Brainstorming (BS)
Causal factors of owner risks and strategies of responding to them	1	Poor communication	Develop a detailed communication plan (identify channels, frequency, and responsibilities)	Develop a communication plan that identifies channels, responsibilities, and timing for each information flow	[47]	✓	
			Adopt regular meetings and progress reports	Addressing issues promptly to ensure alignment by holding regular meetings with all parties	[48]		
			Use digital information management platforms such as BIM 360	Share documents and updates in real time using specialized systems like Aconex or Procore	[49]		
	2	Lack of owner experience	Hiring a Project Management Consultant (PMC)	Providing external expertise to fill gaps	[50]	✓	
			Training the owner on project management basics	Documenting decisions and lessons learned	[47]		✓
			Developing a procedures manual for the owner to make decisions	Delegating technical tasks to an integrated contractor reduces administrative burdens		✓	✓
	3	Site readiness issues	Conducting preliminary field surveys	Evaluate infrastructure and facilities before contracting		✓	
			Best practices for site infrastructure planning	Divide the project into ready-made site sections	[33]		
			Guidelines for assessing site readiness during the preliminary phase or preparing a preliminary field feasibility study	Leverage the contractor's expertise in determining requirements	[50]	✓	✓
	4	Unexpected site conditions	Conducting detailed geotechnical studies (extensive exploration)	Perform advanced geotechnical studies (e.g., ground-penetrating radar (GPR) or core drilling) and increase the sampling density above the applicable standard minimum, using the specific increment determined by project risk and site variability, rather than a fixed percentage such as 20%, and prepare a 3D ground model (Ground Model).	[35, 51]		
				Geotechnical Standards (site characterization framework: American Society for Testing and Materials (ASTM) D420; GPR-specific guidance: ASTM D6432)			
				Financing unforeseen costs, or legally determining the allocation of risk between the owner and contractor			✓
	5	External risks	Including “unforeseen circumstances” clauses in the contract	Using risk simulation software such as @RISK or Monte Carlo Simulation to predict the impact of unforeseen circumstances on schedule and costs. (Conditions of Contract of the International Federation of Consulting Engineers (FIDIC) Red Book)	[7, 52]		✓
			Using Building Information Modelling (BIM) for scenario simulation	Protecting parties from uncontrollable events (wars, disasters)			✓
			Develop an emergency plan that includes natural disasters and political unrest	Planning ahead for alternatives			✓
			Diversify suppliers to avoid dependence on a single source				

Causal factors of consultant risks and strategies and responding methods	1	Design flaws/ inefficiencies	Continuously monitor economic and legal indicators	Transferring financial risks to specialized entities	✓	✓
			Implementing multiple technical design reviews and independent auditing	Implementing periodic design reviews at multiple stages (e.g., 30%, 60%, 90%), conducted by the internal design team and independent external reviewers, with the participation of all parties, to detect errors early and identify them before implementation [18]		
			Using BIM and advanced analysis	Using BIM to detect conflicts, simulate performance, and analyze structural details to improve design accuracy and efficiency prior to design [53]		
			Independent review by selecting and hiring an experienced and reputable design consultant	Using experts in specific disciplines (such as structure or soil) to review the design and address gaps [27]	✓	✓
			Develop a detailed communication plan (identifying channels, frequency, and responsibilities)	Develop a communication plan that identifies channels, responsibilities, and timing for each information flow [47]	✓	
			Hold regular meetings and progress reports	Addressing issues promptly to ensure alignment by holding regular meetings with all parties [30]		
			Use digital information management platforms such as BIM 360	Share documents and updates in real time using specialized systems like Aconex or Procore [49]		
			Requirements tracking system	Assign a person/team to monitor regulatory updates and ensure their implementation in designs and approvals		✓
			Appointment of a compliance officer (compliance review)	Conduct a weekly internal audit using the Plan-Do-Check-Act (PDCA) methodology, documenting the results in traceable records to avoid legal accountability [54]	✓	
			Contractual sanctions	Include penalty clauses in contracts in accordance with FIDIC conditions, such as deducting 0.1% of the contract value per day for each violation, along with compensation for the resulting damages [34, 52]	✓	
			Quarterly technology assessment or semi-annual technology survey	Conduct periodic evaluations of the technologies used and analyze the latest technologies (design, systems, materials, software, and equipment) and their impact on the project	✓	✓
			Adoption of open and scalable standards and systems	Select materials and systems that rely on open standards that can be expanded or updated with relative ease in the future		✓
			Update clauses in consultant contracts or adoption of a gradual update model	Include conditions in the contract that require the consultant to use modern technologies and specify a mechanism for development when necessary [55]		✓

1	Cost overruns	Detailed cost analysis	Implementing the Earned Value Management (EVM) system to compare actual performance with plan is essential. It detects deviations early through the Cost Performance Index (CPI) and Schedule Performance Index (SPI) indicators and allows for immediate course correction	[56]		✓
		Contingency reserve	A management reserve is reserved (industry practice often quotes 5 to 10 per cent of total cost, but this is convention rather than empirically validated; more rigorous approaches set the reserve based on statistical confidence intervals for the project (such as the P80 to P95 range) rather than a flat percentage) and is used only for unidentified (“unknown”) risks after approval by senior management	[57]	✓	✓
		Fixed price contracts (lump sum)	Transferring cost risk to the contractor through fixed-price contracts with strict controls, while complying with the application of quantitative and qualitative risk analysis to determine the true profit margin	[58]		
		Requirements tracking system	Assign a person/team to monitor regulatory updates and ensure their implementation in designs and approvals		✓	
2	Non-compliance with regulations	Appointment of a compliance officer (compliance review)	Conduct a weekly or monthly internal continuous audit using the quality Plan-Do-Check-Act (PDCA) methodology, documenting the results in traceable records to avoid accountability	[54]		
		Contractual sanctions	Add conditions of penalties/wasting and being fit to actual FIDIC, based at 0.1% per day from the value of the contract in case of violation to be followed by compensation for damages ensued	[34, 52]		
		Application scaling (piloting)	Applying the Technology Readiness Level (TRL) model to test tools (such as BIM and the Internet of Things (IoT)) in a limited work environment before scaling up, while measuring the financial return on investment (ROI)	[43, 55]		✓
3	Failure to adopt technology	Technical partnerships	Conclude memoranda of understanding (MoUs) with universities or specialized companies to provide ongoing technical support, and implement knowledge transfer through intensive workshops		✓	
		Adoption incentives	Building a rewards design system based on quantitative and qualitative indicators such as the time/cost reduction ratio resulting from the use of modern technologies, for the purpose of motivating contractors through rewards		✓	✓

Causal factors of contractor risks and strategies and responding methods	4	Inflation/price fluctuations	Proactive purchasing	Apply predictive analytics using economic indicators (such as the Construction Materials Price Index) to stockpile critical materials ahead of anticipated price increases	[59]	✓	
			Price adjustment items	Include a Price Adjustment Formula in contracts that links material prices to official indicators (such as the CPI), with quarterly reviews	[52]		
			Multiple suppliers	Adopt a flexible sourcing strategy by qualifying three or more suppliers for each material, using a Strengths, Weaknesses, Opportunities, and Threats (SWOT) analysis to assess each supplier's capabilities		✓	✓
			Conducting detailed geotechnical studies (extensive exploration)	Performing detailed geotechnical investigation (e.g., boreholes and ground-penetrating radar) whose extent is scaled to the project risk and site variability before construction starts	[35, 51]		
	5	Unforeseen site conditions	Including "unforeseen circumstances" clauses in the contract	Allocate the financial consequences of unforeseen ground conditions between the owner and the contractor through explicit contract clauses (e.g., the relevant FIDIC Red Book provisions)	[52]		
			Using Building Information Modelling (BIM) for scenario simulation	Using risk simulation software such as @RISK or Monte Carlo Simulation to predict the impact of unforeseen conditions on schedule and costs. (Conditions of Contract (FIDIC) Red Book)	[7, 52]		
			Predictive maintenance	It is necessary to install Internet of Things (IoT) devices to collect real-time performance data, actual cost and work quality, and analyze them using artificial intelligence systems to detect failure patterns before they occur, and prioritize correct performance	[44]		✓
	6	Equipment failure	Guaranteed maintenance contracts	Select Total Maintenance Contracts with suppliers that include provisions for immediate replacement and downtime compensation, with payment tied to equipment uptime indicators	[47]		
			Spare equipment	Apply the redundancy principle by providing replacement units for critical equipment (such as power generators), while calculating the optimal cost using cost-benefit analysis (CBA)			✓
			Pre-submission	Applying the Critical Path Method (CPM) technique to identify the critical period and submit it three or six months in advance, including the Environmental Impact Assessment (EIA)	[47]	✓	✓
	7	Delay in permits	Government relations coordinator	Appoint a stakeholder manager to build a network with government agencies, using the Power/Interest Grid to identify those influencing the approval process	[60]	✓	✓

Alternative plans (fast-tracking)	Apply a fast-tracking approach to design, such as starting site work before completing finishing permits, with performance bonds to cover legal risks	[61, 62]	✓
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Note: BS and EI denote the source-identification methods; it refers to the sources from which the risk factors listed in Table 3 were identified also in section 2.3.

3.5 Responses to Causal Risk Factors for Project Owners, Consultants, and Contractors

This section includes response identification strategies collected and classified through various research sources, EI, and BS. Table 16 illustrates the response strategies to causal factors to mitigate their risks and impact on the affected factors. Note that the table includes the causal factor, the main response strategies, and an explanation of the response methodologies.

4 Conclusions and Recommendations

This study indicated the necessity of adopting a proactive approach to identifying and analyzing the root causes and risks of key stakeholders involved in project implementation. This study employed a framework integrating fuzzy logic and DEMATEL techniques to understand the complex causal relationships among stakeholders involved in project execution. Primary and secondary risk factors were examined, thus leading to the following conclusions.

4.1 Conclusions

4.1.1 Results of the fuzzy DEMATEL study

The following direct findings were obtained from the FD analysis.

First, according to the model, the Consultant ($D - R = -0.84$) is the main receiver of transmitted risk, while the Owner ($D - R = +0.739$) and the Contractor ($D - R = +0.108$) are causative generators of risk (Table 12).

Second, four cross-party transmission linkages over the significance threshold are identified by the crisp total-relation matrix (Table 11): Owner $\rightarrow$ Consultant, Owner $\rightarrow$ Contractor, Contractor $\rightarrow$ Consultant, and Contractor $\rightarrow$ Owner. The consultant does not maintain any outgoing link above the threshold.

Third, within the owner matrix, the sub-factors with the highest prominence were financial instability (RC1, $D + R = 5.993$) and stakeholder conflicts (RC8, $D + R = 5.962$), while poor communication (RC5, $D + R = 5.756$) ranked fourth. Because the owner, consultant, and contractor sub-matrices were computed separately, $D + R$ values are comparable only within each stakeholder matrix and are not compared across matrices here.

Fourth, the owner-side variables classified as causal (positive $D - R$)—poor communication (RC5), site readiness issues (RC7), site condition uncertainties (RC9), and external risks (RC10) are those most closely linked to performance concerns on the consultant and contractor sides (Section 3.2).

4.1.2 Managerial implications and interpretation

Beyond what the model specifically evaluates, these findings have two more general implications for practice.

First, mitigation resources are likely to be used most effectively if they are focused on the Owner's decision-making processes rather than being distributed equally among the three stakeholders because the Owner is both the strongest causal source and a common origin point for risk that reaches the other two parties.

Second, although the FD results identify communication and site-condition-related factors as net causal factors (positive $D - R$) within the owner matrix, rather than as the factors with the highest prominence. It has been discovered that in practice these effects are often exacerbated by inadequate planning and coordination; this connection is presented as an interpretive observation for practitioners, rather than a consequence of the causal analysis itself. This observation is consistent with the general project management literature rather than as a direct output of the model.

4.2 Recommendations

Some preventive measures could be suggested to prevent possible occupational hazards considering the result. First, emphasis should be placed on the cause group criterion as it drove the impact group criteria. It is much more challenging to prioritize the criteria of the cause group than those of the impact.

- Enhance communication management between the parties: To guarantee transparency and cut down on decision-making delays, create digital platforms for information sharing between the Owner, Consultant, and Contractor. Organize regular coordination meetings with all parties' representatives to go over possible dangers beforehand.
- Enhance the consultant's function as a successful mediator by involving them early in the planning stage to prevent specification differences. Provide consultants with explicit performance requirements, such as timely response and dispute resolution.

4.3 Study Limitations and Future Research

This study has intrinsic limitations despite the organized insights it offered. First, notwithstanding their specialization, the panel of nine experts that produced the results represents a particular rather than a general viewpoint. Second, because the analysis is conducted in a particular nation, the identified causal weights might change in other regulatory or economic contexts. Lastly, the conclusions are based on expert opinion and are not supported by empirical evidence from finished building projects. To confirm the anticipated causal relationships in practical contexts, future research should concentrate on using this paradigm in longitudinal case studies.

Author Contributions

Conceptualization, Y.S.N. and A.R.R.; methodology, Y.S.N. and A.R.R.; software, A.E.H. and N.A.R.M.; formal analysis, Y.S.N.; investigation, A.E.H. and N.A.R.M.; resources, A.R.R.; data curation, Y.S.N. and N.A.R.M.; writing, A.E.H. and Y.S.N.; writing—review and editing, A.R.R. and N.A.R.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

All data and equations supporting our research results are included within the article.

Conflicts of Interest

The authors declare no conflicts of interest.

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